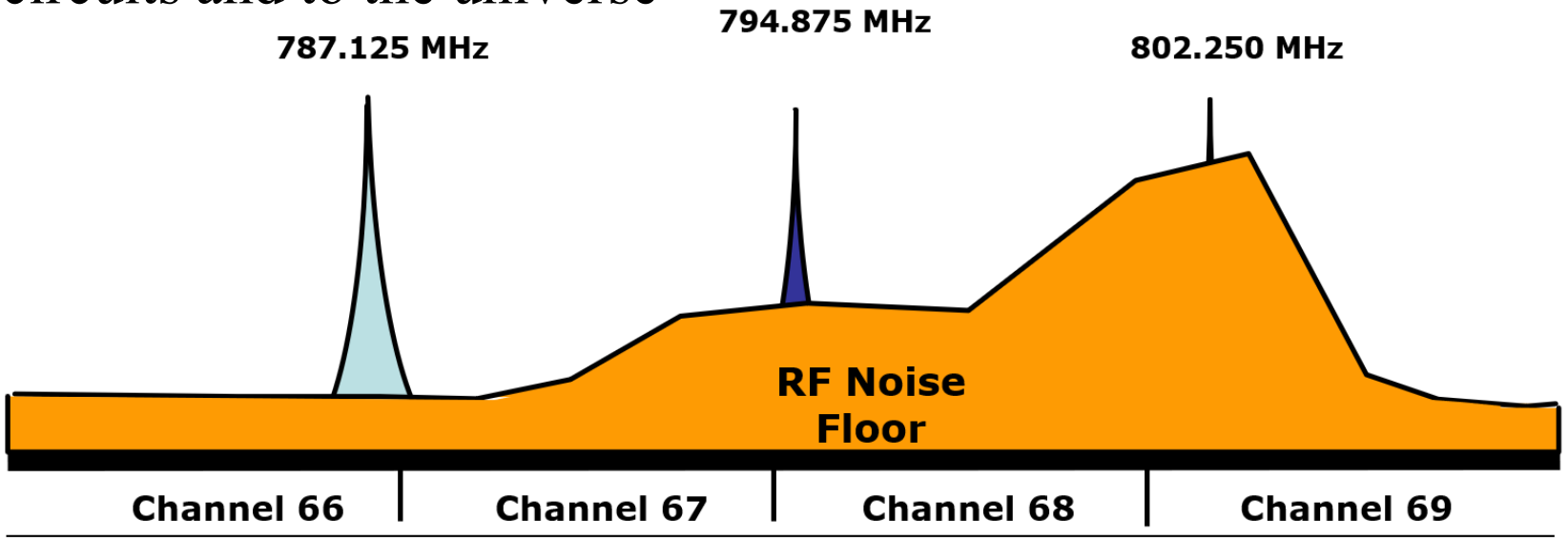
EEME 30004
Communications & Networks

Noise

Dr Sean Gu

Noise

- **Undesired** in RF and microwave communications, radar, and remote sensing systems
- Anything added to an electrical signal that **makes the extraction of the contained information more difficult**
- Comes from the discrete nature of electrons (**thermal vibrations in any component at a temperature above absolute zero**), such as the flow of charges or holes in an electron tube or solid-state device, propagation through the ionosphere or other ionized gas
- **Can be minimized, but never eliminated**, because it is intrinsic to components, to the operation of circuits and to the universe

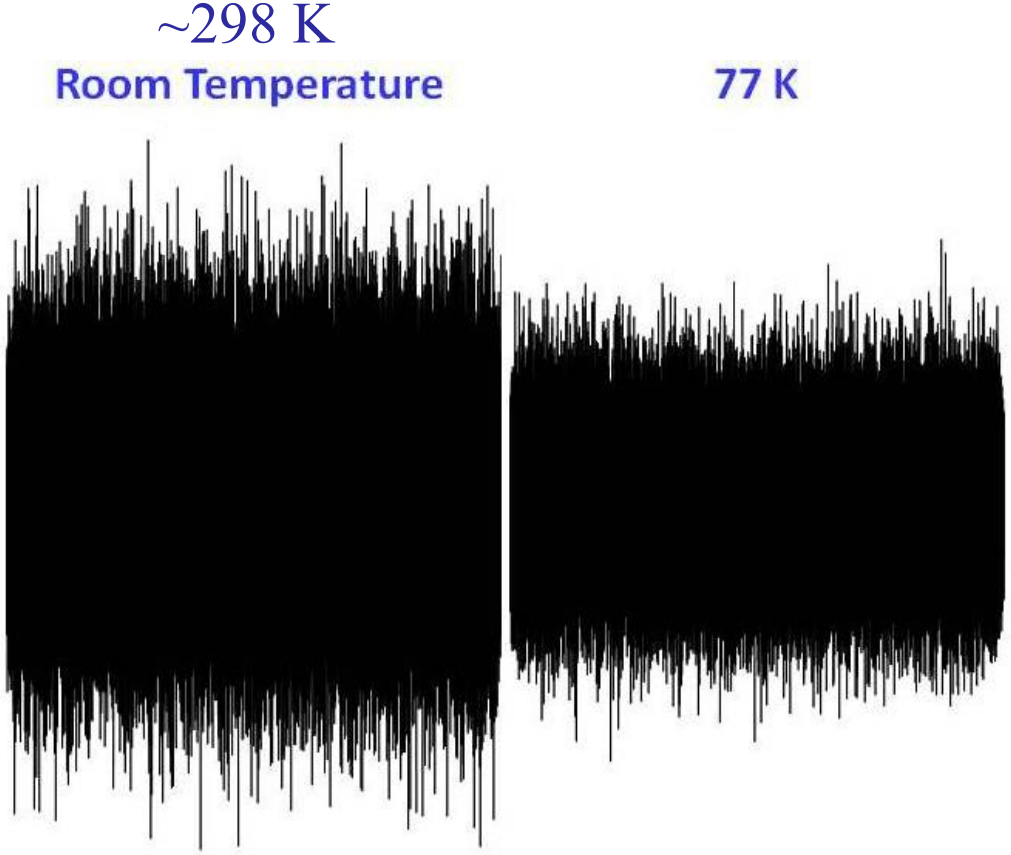


Noise

Type of noise: (internal)

- ***Thermal noise:*** the most basic type of noise, being caused by thermal vibration of bound charges. It is also known as *Johnson* or *Nyquist* noise.
- ***Shot noise:*** due to random fluctuations of charge carriers in an electron tube or solid-state device.
- ***Flicker noise:*** occurs in solid-state components and vacuum tubes and its power varies inversely with frequency, and so is often called $1/f$ - noise.
- ***Plasma noise:*** caused by random motion of charges in an ionized gas, such as a plasma, the ionosphere, or sparking electrical contacts.
- ***Quantum noise:*** results from the quantized nature of charge carriers and photons; it is often insignificant relative to other noise sources.

thermal noise in nuclear magnetic resonance data

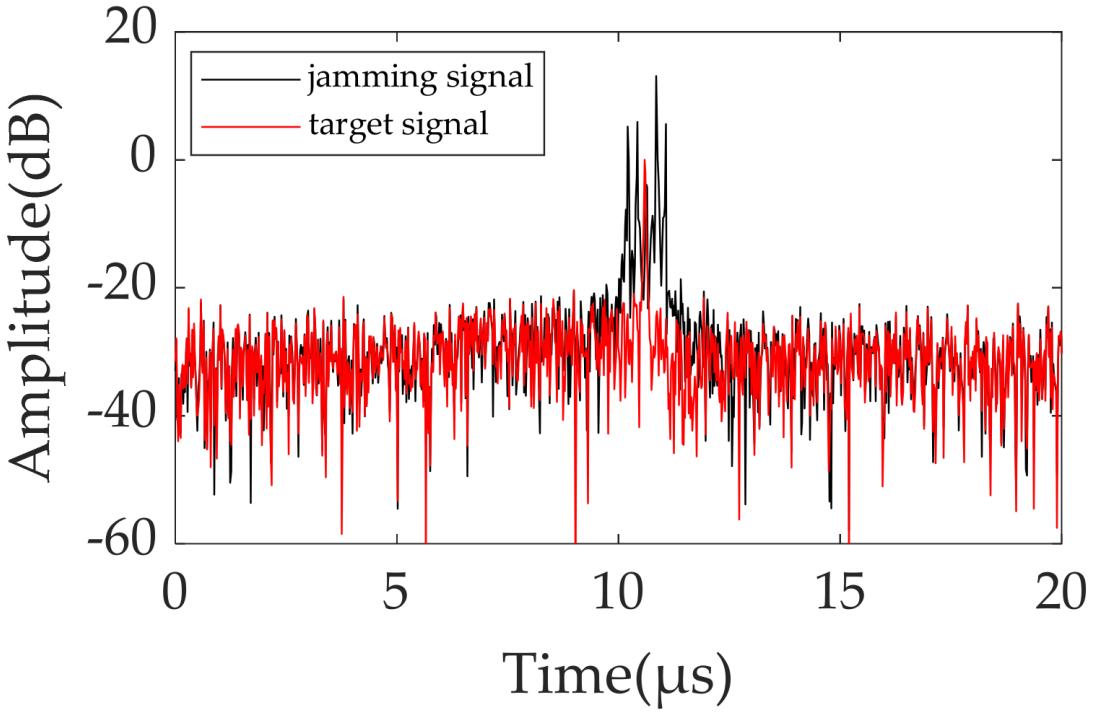


Noise

Type of noise: (external)

- Thermal noise from the ground
- Cosmic background noise from the sky
- Noise from stars (including the sun)
- Lightning
- Gas discharge lamps
- Radio, TV, and cellular stations
- Wireless devices
- Microwave ovens
- Deliberate jamming devices

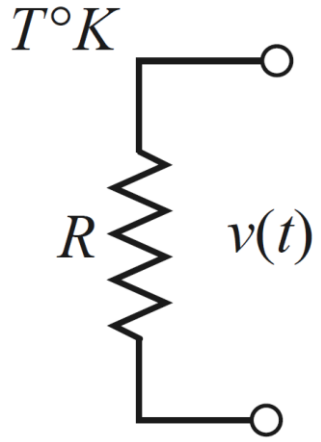
intentional jamming signals



Noise

Thermal noise (*Johnson or Nyquist noise*):

a resistor R at a physical temperature of T degrees kelvin (K).



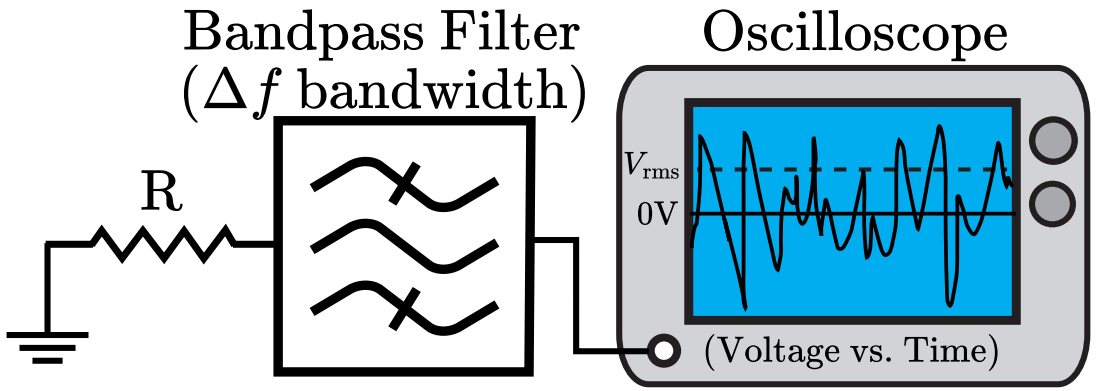
The electrons in the resistor are in random motion, with a kinetic energy that is proportional to the temperature. These random motions produce small, random voltage fluctuations at the resistor terminals. This voltage V_n has a zero average value but a nonzero root mean square (rms) value:

$$V_n = \sqrt{4kTBR}$$

k: Boltzmann constant (1.38×10^{-23} J/K)

T = absolute temperature [K] (273.15 +T ºC)

B = considered bandwidth [Hz]

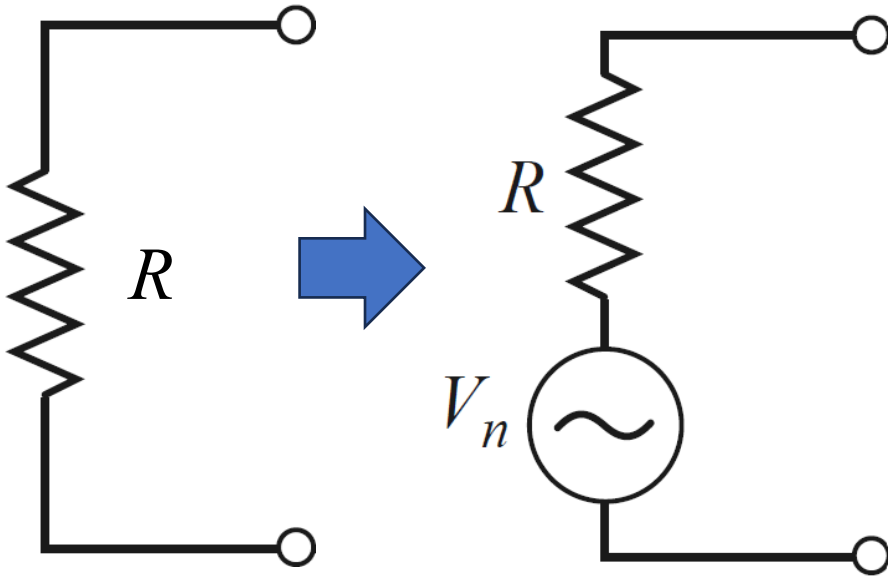


Noise

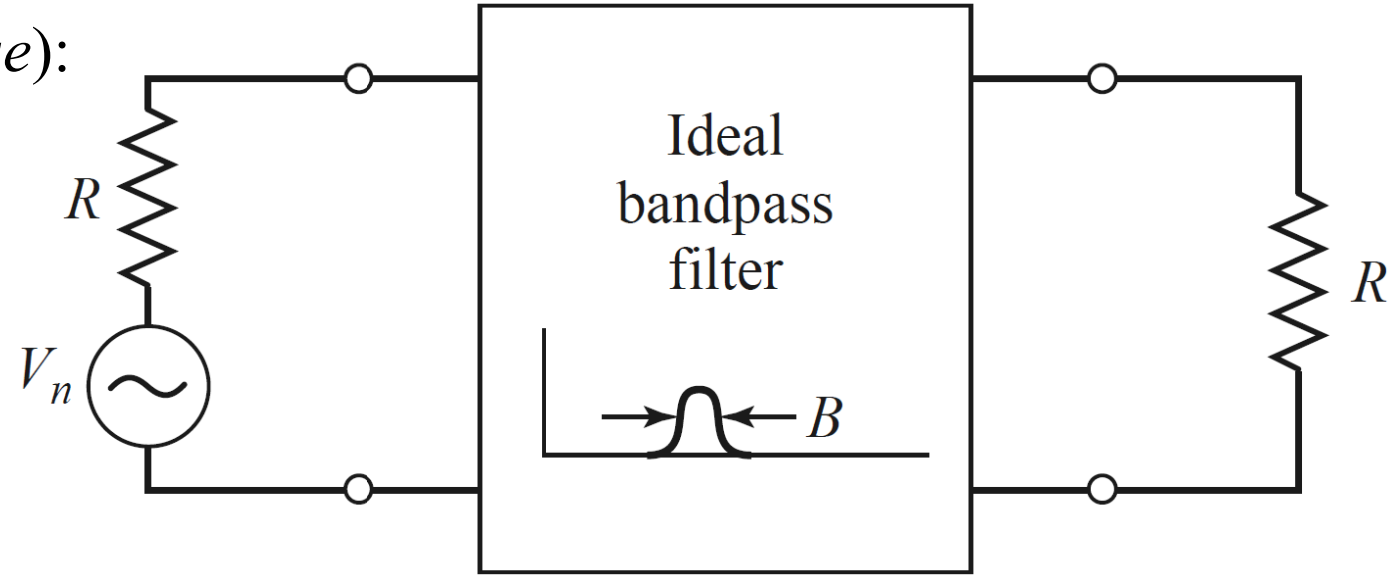
Thermal noise (*Johnson or Nyquist noise*):

this resistor R is a source of thermal noise and can be modeled by a noiseless resistor and a generator with a voltage:

$$V_n = \sqrt{4kTBR}$$



equivalent circuit



Connecting this noisy resistor with a load resistor R would result in maximum power transfer.
The power delivered to the load in a bandwidth B can be written as:

$$P_n = \left(\frac{V_n}{2R}\right)^2 R = \frac{V_n^2}{4R} = kTB$$

Noise

University of BRISTOL
School of Electrical, Electronic and Mechanical Engineering

Thermal noise (*Johnson or Nyquist noise*):

$$V_n = \sqrt{4kTBR}$$

Rule of thumb: 50 Ω resistor at 1 Hz bandwidth corresponds to 1 nV noise at room temperature.

Note: the source is not sinusoidal and has no polarity.

$$P_n = \left(\frac{V_n}{2R}\right)^2 R = \frac{V_n^2}{4R} = kTB$$

As $B \to 0$, $P_n \to 0$. This means that systems with smaller bandwidths collect less noise power.
As $T \to 0$, $P_n \to 0$. This means that cooler devices and components generate less noise power.
As $B \to \infty$, $P_n \to \infty$. This is the so-called *ultraviolet catastrophe*, which does not occur in reality.

Noise

University of BRISTOL
School of Electrical, Electronic and Mechanical Engineering

Equivalent noise temperature T_e :

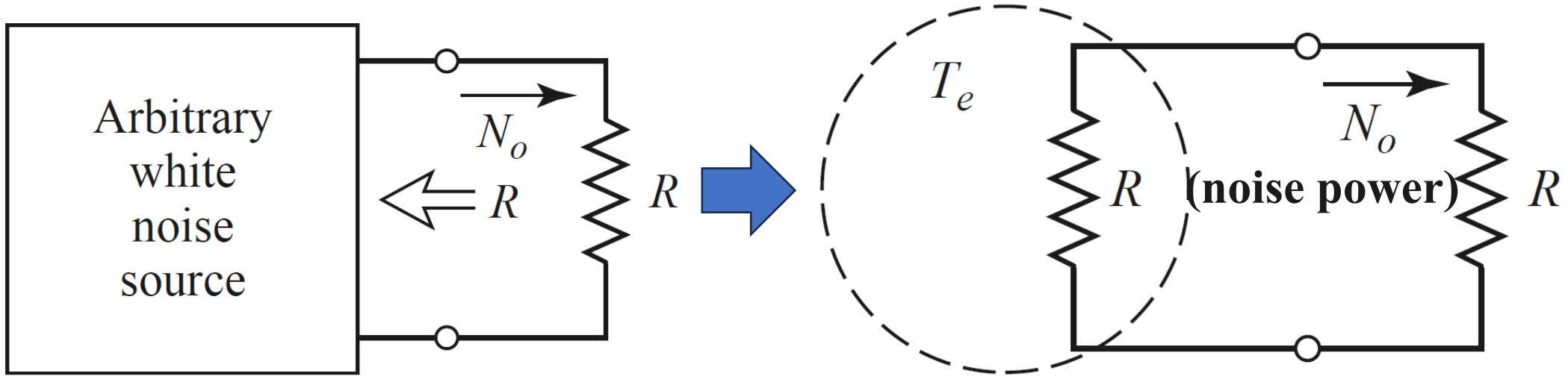
arbitrary source of noise (thermal or nonthermal) is "white" (equal intensity at different frequencies), meaning the noise power is not a strong function of frequency over the bandwidth of interest.

modeled as an equivalent thermal noise source. characterized with an ***equivalent noise temperature*** T_e:



$$N_o = \left(\frac{V_n}{2R}\right)^2 R = \frac{V_n^2}{4R} = kTB$$

$T_e = ?$



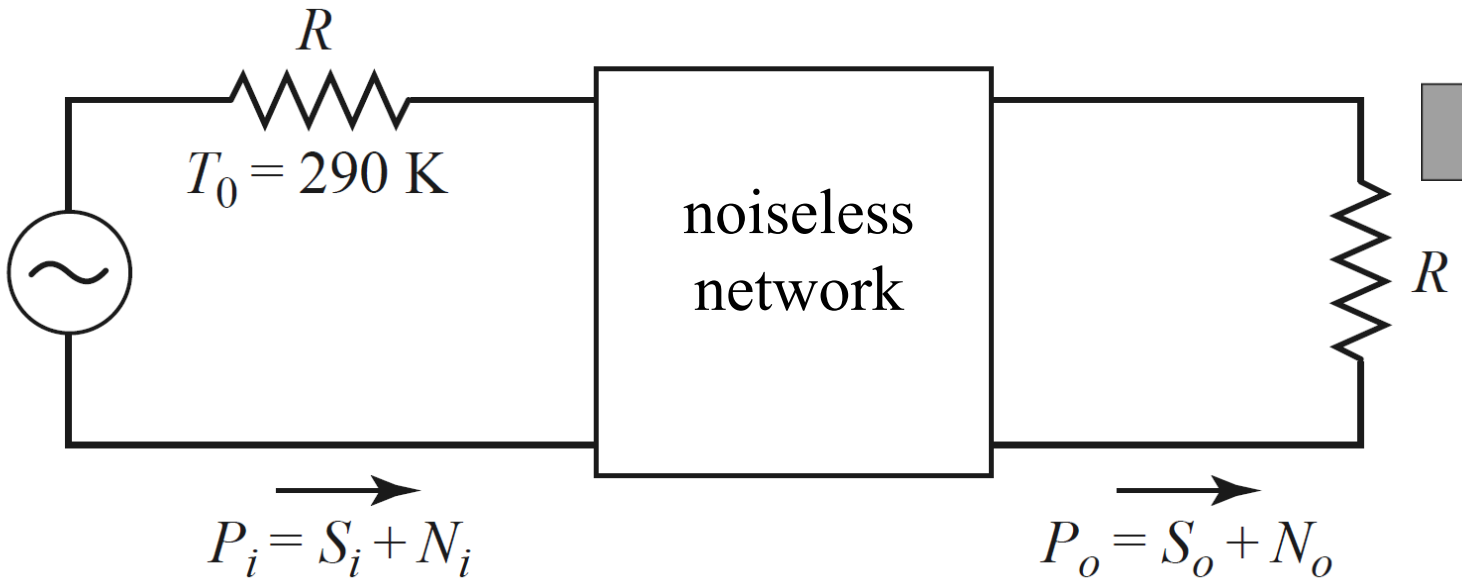
Noise

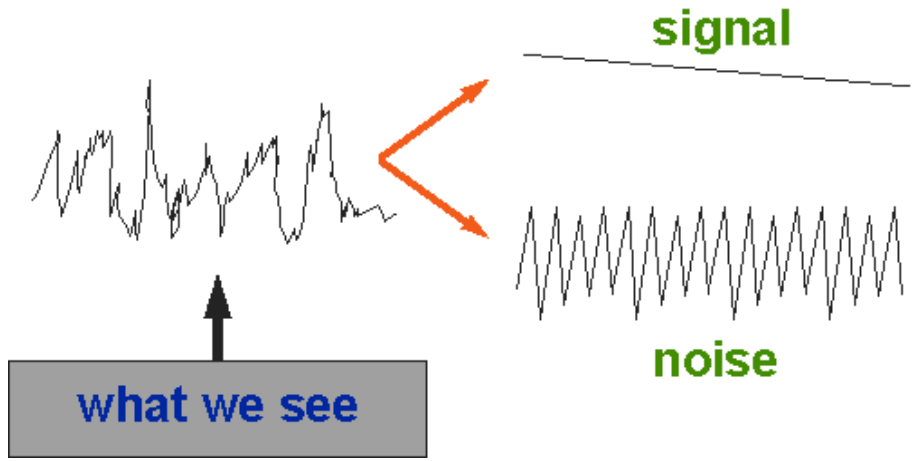
University of BRISTOL
School of Electrical, Electronic and Mechanical Engineering

Signal-to-Noise Ratio (SNR): ratio of desired signal power to undesired noise power

SNR = desired/undesired = signal/noise = S/N

SNR (dB) = $10 \log_{10} (S/N)$





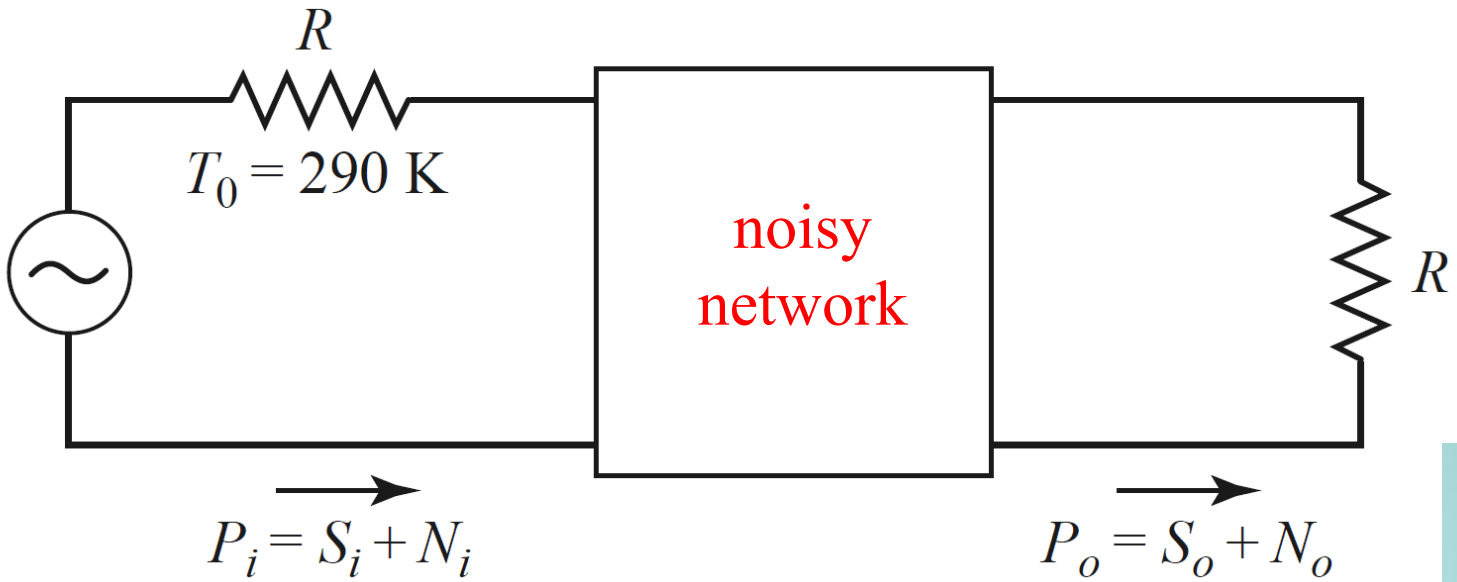
noiseless network: both signal and noise power will be attenuated or amplified by the same factor, so SNR remains the same.

S_i and N_i are input signal and noise power; S_o and N_o are output signal and noise power.

Noise

University of BRISTOL
School of Electrical, Electronic and Mechanical Engineering

Signal-to-Noise Ratio (SNR): ratio of desired signal power to undesired noise power



S_i and N_i are input signal and noise power; S_o and N_o are output signal and noise power.

$S_i \Rightarrow S_o$ $\quad N_i \Rightarrow N_o$



$$SNR_o \begin{cases} > \\ < \\ = \end{cases} SNR_i$$

Noise

University of BRISTOL
School of Electrical, Electronic and Mechanical Engineering

Noise figure F quantifies the deterioration of the signal-to-noise ratio (SNR) due to the insertion of a network into the system:

$$F = \frac{SNR_i}{SNR_o} = \frac{S_i/N_i}{S_o/N_o} = \frac{N_o}{A_p \times N_i} \qquad A_p = \text{power gain}$$

- Ideally, we have $F = 1$, if the stage does not add any noise.
- In practice, $F > 1$.
- A higher F indicates a deterioration in the *SNR* between the input and the output of the inserted stage.

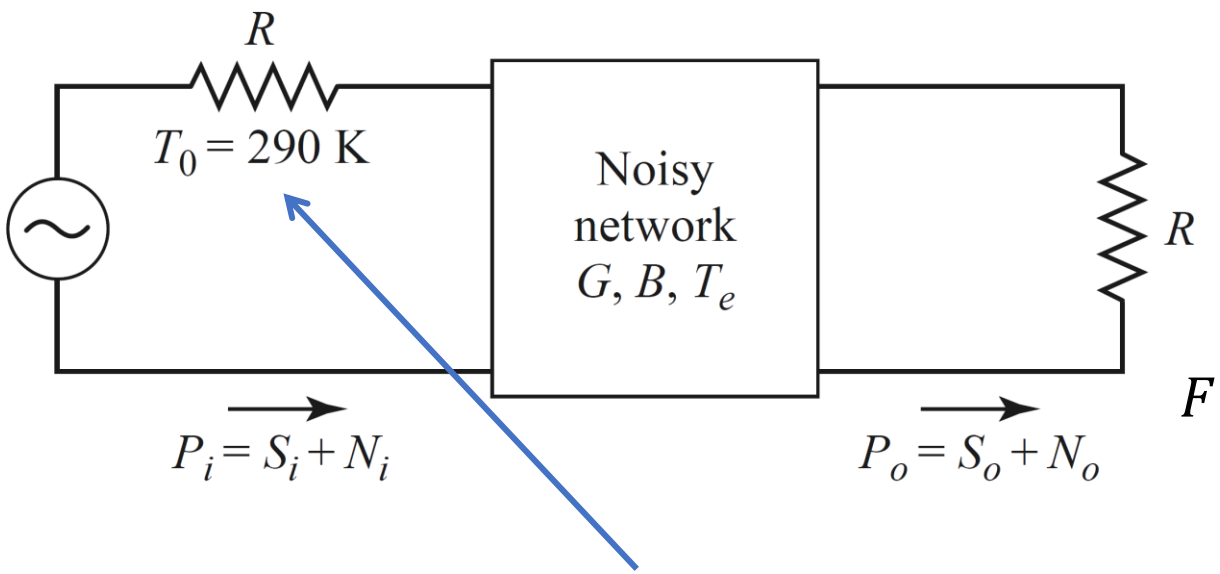
$$F(\text{dB}) = 10 \log\left(\frac{S_i/N_i}{S_o/N_o}\right)$$

- Ideally, we have $F(\text{dB}) = 0$ dB, if the stage does not add any noise.
- In practice, $F(\text{dB}) > 0$ dB.
- A higher value $F(\text{dB})$ indicates a deterioration of the *SNR* between the input and the output of the inserted stage.

Noise

University of BRISTOL
School of Electrical, Electronic and Mechanical Engineering

Noise figure F & equivalent noise temperature T_e:



noise power at the input: $N_i = kT_0B$

noise power at the output: $N_o = kGB(T_0 + T_e)$

signal power at the output: $S_o = GS_i$

$$F = \frac{SNR_i}{SNR_o} = \frac{S_i/N_i}{S_o/N_o} \Rightarrow F = \frac{S_i/kT_0B}{GS_i/kGB(T_0 + T_e)}$$

$$F = 1 + \frac{T_e}{T_0} \geq 1$$

$$T_e = (F - 1)T_0$$

Noise figure is defined for a matched input source, and for a noise source equivalent to a matched load at temperature $T_0 = 290$ K.

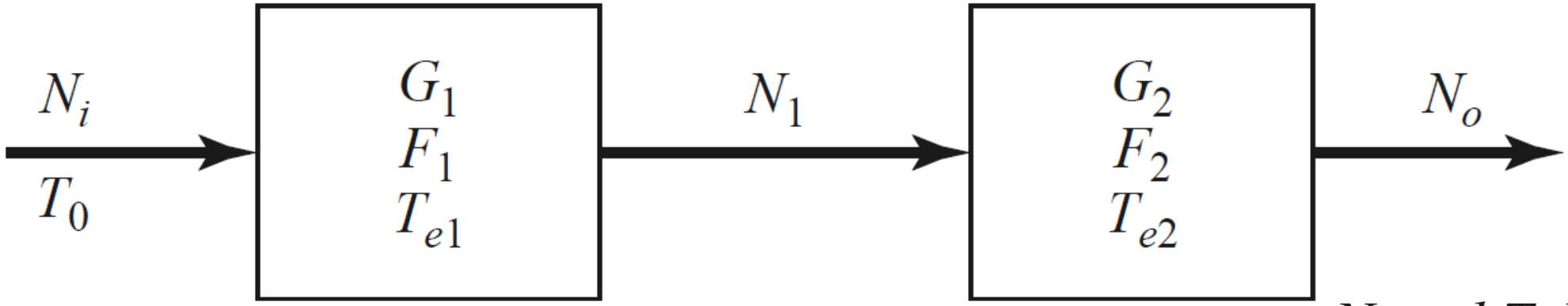
Noise figure and equivalent noise temperatures are interchangeable characterizations of the noise properties of a component.

Noise

University of BRISTOL
School of Electrical, Electronic and Mechanical Engineering

Noise figure of a cascade system:

G: gain, F: noise figure, T_e: equivalent noise temperature



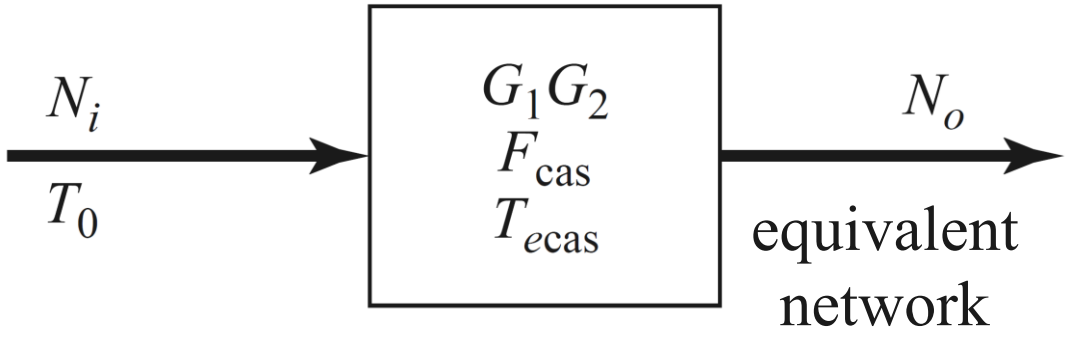
$N_i = kT_0B$ (input)

noise power at the output of the first stage: $N_1 = G_1kT_0B + G_1kT_{e1}B$ (network 1)

noise power at the output of the second stage: $N_o = G_2N_1 + G_2kT_{e2}B$ (network 2)

$$N_o = G_1G_2kB\left(T_0 + T_{e1} + \frac{1}{G_1}T_{e2}\right)$$

$$N_o = G_1G_2kB(T_{cas} + T_0)$$

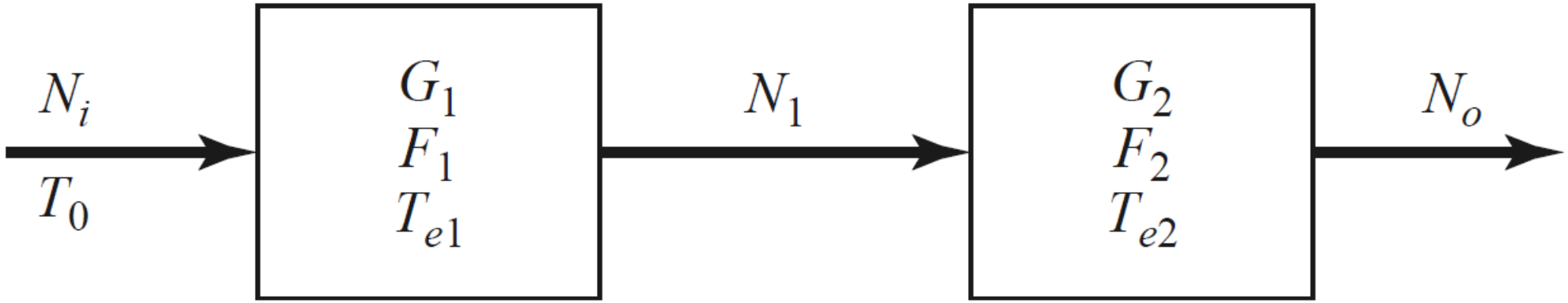


Noise

University of BRISTOL
School of Electrical, Electronic and Mechanical Engineering

Noise figure of a cascade system:

G: gain, F: noise figure, T_e: equivalent noise temperature



noise power at the output of the second stage: $N_o = G_1 G_2 k B (T_{\text{cas}} + T_0)$

$$N_o = G_1 G_2 k B \left(T_0 + T_{e1} + \frac{1}{G_1} T_{e2} \right)$$

noise figure of the cascade system:

$$F_{\text{cas}} = F_1 + \frac{1}{G_1}(F_2 - 1)$$

$$T_e = (F - 1) T_0$$

noise temperature of the cascade system:

$$T_{\text{cas}} = T_{e1} + \frac{1}{G_1} T_{e2}$$

Noise

University of BRISTOL
School of Electrical, Electronic and Mechanical Engineering

Noise figure of a cascade system:

Friis formula for noise: makes it possible to determine the noise figure of a system with several stages in cascade

$$F_{\text{cas}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots$$

F_{cas} = noise figure of a cascade system
F_i = noise figure of stage i
G_i = power gain of stage i

linear scale If given values in dB, convert them into linear scale first!

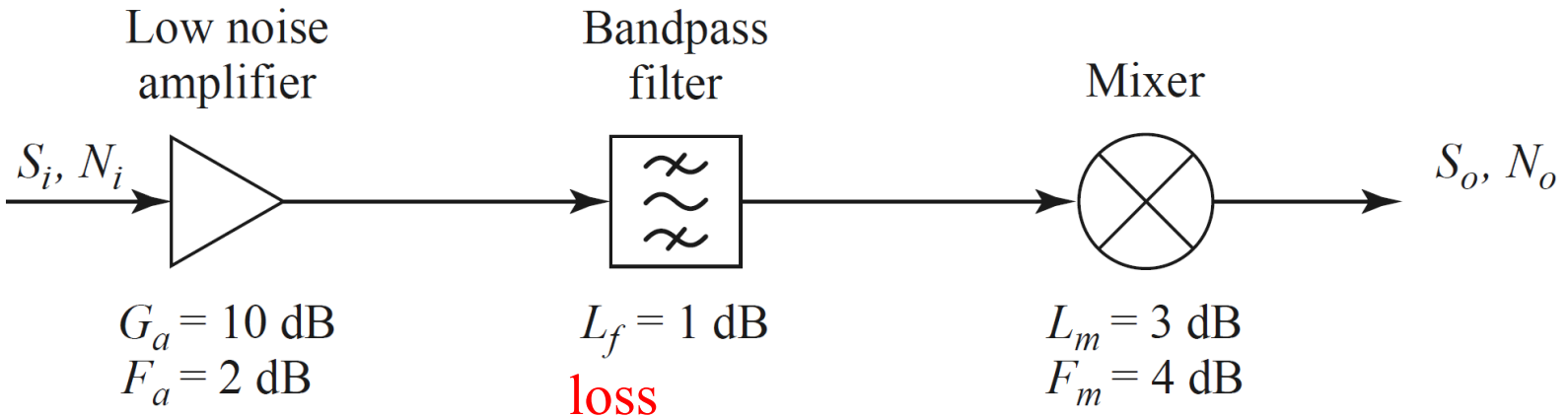
Stages do not all contribute the same. The ***first stage*** is the most important!

$$T_{\text{cas}} = T_{e1} + \frac{T_{e2}}{G_1} + \frac{T_{e3}}{G_1 G_2} + \cdots$$

T_{cas} = noise temperature of a cascade system
T_{ei} = equivalent noise temperature of stage i

Noise

Noise figure of a cascade system:





Compute the overall noise figure of this subsystem.

loss and gain relationship: $L = 1/G$ linear

$L = -G$ in dB

Convert from dB to numerical values:

loss and noise figure relationship: $F = 1 + (L - 1)\frac{T}{T_0}$

$F = L$ if room temperature

Noise

Some minimum values of SNR:

- 3 dB is the minimum detectable for any system
- 10 dB is the minimum for detection for the AM receiver
- 12 dB is the minimum for detection for the FM receiver
- 40 dB is the minimum for detection for the TV receiver

In most systems, noise is primarily caused by:

- The noise present in the input signal
- The noise caused by the first two stages of the system

$$F_{\text{cas}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots$$

Questions?

Sean Gu
sean.gu@bristol.ac.uk